B. Tech. Examination 2021
Computer Science Engineering (CSE)
(Odd Semester Regular and Supplementary)
Microprocessor and its Application (ECEUGOE04)

F.M: 80 **Time: 3.00 Hrs**

GROUP A **1X10**
(Multiple-Choice Questions)
Answer any ***ten***

1. **(i)** SP register holds the

a) Base address of stack b) Address of stack top.
c) Address of the instruction to be fetched. d) None of these.

(ii) 8085 microprocessor is an example of

a) Harvard architecture b) Von-Neumann architecture
c) Both (a) and (b) d) None of these.

(iii) A single instruction to clear the lower four bits of the accumulator in 8085 microprocessor is

a) XRI $0F_H$ b) ANI $F0_H$
c) ANI $0F_H$ d) XRI $F0_H$

(iv) The control word to set PC7 of 8255 in BSR mode is

a) $0C_H$ b) $0D_H$
c) $0E_H$ d) $0F_H$

(v) INX instruction affects

a) All flags. b) All flags except zero flag.
c) All flags except carry flag. d) No flags

(vi) How many T-states are required to execute DAA instruction?

a) 10 b) 13
c) 4 d) 6

(vii) The memory capacity of the 8085 microprocessor is

a) 64K b) 32K
c) 32MB d)64MB

(viii) $(AA)_H - (11)_H = (?)_H$

a) 00 b) 99
c) A0 d)09

(ix) Which one of the following is the software interrupt of 8085 microprocessor?

a) RST7.5 b) RST0
c) TRAP d) INTR

(x) The instruction: EXHG exchanges the contents of

a) Acc & H b) BC-pair & HL- pair

c) DE-pair & HL- pair d) HL-pair and memory

(xi) Instruction queue of 8086 consists of

a) 4 data b) 6 data

c) 8 data d) 16 data

(xii) The address bus of 8085 microprocessor is

a) Unidirectional b) Bidirectional

c) Unidirectional as well as Bidirectional d) None of these

GROUP B

(Long answer type questions)

*Answer any **seven***

2. **a)** What is the need to demultiplexing the address/data bus? How is it done? (5+5)

b) Explain the following signals of 8085microprocessor:

a) READY b) S0 & S1 c) HOLD d) ALE

3. **a)** Why are PC and SP 16-bit register? (3+7)

b) Can the same port address be assigned to two I/O devices if interfaced in peripheral mapped I/O? If yes, how and if no, why? Differentiate between Memory mapped I/O and Peripheral mapped I/O.

4. Write a program to convert a hexadecimal number stored in memory location 8200_h to its corresponding ASCII. Store the result from 8300_h. (10)

5. **a)** CALL and RET are the only instructions of 8085 that uses subroutine. Comment on this statement. (5+5)

b) Explain the instruction of 8085 that uses auxiliary carry flag. Write a program to clear the flags of 8085.

6. Interface the 4KB RAM, 2 KB EPROM and 4 KB EPROM memory devices in 8085 microprocessor in absolute decoding method. (10)

7. **a)** Explain the role of stack in executing a subroutine. (4+6)

b) Draw the timing diagram for the instruction INR M stored in memory location 9550_H.

8. **a)** Draw and explain the control word format of 8255. Write a control word for port A and port C as an input port and port B as an output port. (6+4)

b) Explain mode 1 operation of 8255 when used as an output port with proper diagrams.

9. **a)** Draw and explain the flag register of 8086 microprocessor (5+5)

b) Explain the process of segmentation of memory in 8086 microprocessor.

10. What is data and address size in 8086? Draw and discuss in brief the BIU and EU of 8086 microprocessor. (10)

11. **a)** Explain the instructions SIM and RIM of 8085 microprocessor. (5+5)

b) What do you mean by maskable and non maskable interrupts of 8085? Write a program to enable all interrupt except RST5.5 interrupt.

12. Write short notes on ***any two*** of the followings: 2X5

(a) 8086 supports pipelining.
(b) The 8251 USART
(c) DMA operation
(d) Addressing modes of 8085